

Chapter B4: Using food and controlling growth

Overview

All living organisms depend on molecules of glucose obtained from photosynthesis (or from biomass obtained through food chains that start with photosynthetic organisms). The glucose is used for cellular respiration and in the synthesis of larger organic molecules used for growth.

From study at earlier Key Stages, learners will be familiar with the reactants and products of cellular respiration. In Topic B4.1 they explore how cellular respiration increases the amount of energy associated with cellular energy stores, in particular molecules of ATP that are essential for many life processes. In Topic B4.2 they consider briefly how we came to know what we do about organelles such as mitochondria, using the context of electron microscopy to illustrate the idea that some scientific

explanations were only developed once a technological development made certain observations possible. Topic B4.3 links growth in multicellular organisms to the division of cells during the cell cycle, and explores the nature of stem cells and the role of cell differentiation. As a development of ideas, learners consider how cancer results from changes in DNA that cause a loss of control of cell division.

The role of plant hormones in controlling plant growth and environmental responses is explored in Topic B4.4, and these ideas applied to use of these hormones by humans to control plant growth to our advantage.

Finally, Topics B4.4 and B4.5 explore the question of whether stem cells should be used to regenerate tissue and treat disease.

Learning about cellular respiration and growth before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- be familiar with the processes of aerobic and anaerobic respiration in living organisms, and fermentation in microorganisms, including word summaries of the reactions
- be able to recall the differences between aerobic and anaerobic respiration in terms of the reactants, products and implications for the organism
- be familiar with the tissues and organs of the human digestive system, including adaptations to function
- understand in simple terms that the human digestive system uses chemicals (including enzymes) to digest food
- appreciate the importance of bacteria in the human digestive system
- know how nutrients and water are transported within animals, including humans.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about cellular respiration and growth at GCSE (9–1)

B4.1 What happens during cellular respiration?

Teaching and learning narrative

Consumers gain biomass from other organisms when they eat them. Some of this biomass is converted into molecules of glucose, the fuel for cellular respiration.

Cellular respiration involves many chemical reactions and makes molecules of ATP. It occurs in the cytoplasm and mitochondria of animal and plant cells, and in the cytoplasm of microorganisms. ATP is required for processes that are essential for life, including breakdown and synthesis of molecules, active transport and muscle contraction.

Aerobic respiration breaks down glucose and combines the breakdown products with oxygen, making water and carbon dioxide (a waste product).

In conditions of low or no oxygen (such as in human cells during vigorous exercise, plant root cells in waterlogged soil and bacteria in puncture wounds) anaerobic respiration occurs. There is a partial breakdown of glucose, producing fewer molecules of ATP. In animal cells and some bacteria, this produces lactic acid (a waste product). In plants and some microorganisms, including yeast, it produces ethanol and carbon dioxide.

Assessable learning outcomes

Learners will be required to:

1. compare the processes of aerobic and anaerobic respiration, including conditions under which they occur, the inputs and outputs, and comparative yields of ATP
2. explain why cellular respiration occurs continuously in all living cells
3. explain how mitochondria in eukaryotic cells (plants and animals) are related to cellular respiration
4. describe cellular respiration as an exothermic process
5.
 - a) describe practical investigations into the effect of different substrates on the rate of respiration in yeast
PAG5
 - b) carry out rate calculations for chemical reactions in the context of cellular respiration
M1a, M1c

Linked learning opportunities

Practical work:

- investigate the amount of energy released from different foods, by burning them under a boiling tube of water where:

$$\text{energy (kJ)} = \text{mass of water (kg)} \times \text{change in temperature (deg C)} \times 4.2 \text{ kJ/kg/deg C}$$
- investigate respiration in microorganisms by collecting CO₂ given off; which substrate works best?

B4.2 How do we know about mitochondria and other cell structures?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Scientific progress often relies on technological developments which enable new observations to be made. The invention of the electron microscope enabled us to observe cell organelles such as mitochondria and chloroplasts at much higher magnification than had previously been possible with light microscopes, and thus to develop explanations about how their structures relate to their roles in cellular processes (1aS3).	1. explain how electron microscopy has increased our understanding of sub-cellular structures
	2. in the context of cells and sub-cellular structures: a) demonstrate an understanding of number, size and scale and the quantitative relationship between units M2a, M2h b) use estimations and explain when they should be used M1d c) calculate with numbers written in standard form M1b

Linked learning opportunities

Ideas about Science:

- explanations about the roles of cell organelles were developed from observations that could only be made using electron microscopy (1aS3)

B4.3 How do organisms grow and develop?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Growth of multicellular organisms involves an increase in the number of body cells. All new cells are created from existing cells when they divide. New body cells are created as part of the cell cycle. During interphase the cell grows larger, the numbers of organelles increase, and each chromosome is copied; then during mitosis the chromosome copies separate, the nucleus divides, and the cell divides to produce two new cells that are genetically identical to one another. Cancer is a non-communicable disease in humans caused by changes in a person's DNA. The changes cause a cell to divide many times by mitosis, which can create a tumour.	1. a) describe the role of the cell cycle in growth, including interphase and mitosis b) describe how to use a light microscope to observe stages of mitosis <i>PAG1</i> ① <i>Learners are not expected to recall intermediate phases</i> 2. describe cancer as the result of changes in cells that lead to uncontrolled growth and division
Gametes are produced by meiosis, a different type of cell division. After interphase (during which the chromosome number has doubled), two meiotic divisions occur. Gametes contain half the number of chromosomes found in body cells (one chromosome from each pair). At fertilisation, maternal and paternal chromosomes pair up, so the zygote has the normal chromosome number.	3. explain the role of meiotic cell division in halving the chromosome number to form gametes, including the stages of interphase and two meiotic divisions ① <i>Learners are not expected to recall intermediate phases</i>
A zygote divides by mitosis to form an embryo. All of the cells in an embryo are initially identical and unspecialised; these are embryonic stem cells, and can become specialised to form any type of cell (differentiation) by switching genes off and on. Most cells in a human embryo become specialised after the eight cell stage. However, some (adult stem cells) remain unspecialised and can become specialised later to become many, but not all, types of cells. In plants, only cells in meristems undergo mitosis, producing unspecialised cells that can develop into any kind of plant cell.	4. describe the function of stem cells in embryonic and adult animals and meristems in plants 5. explain the importance of cell differentiation, in which cells become specialised by switching genes off and on to form tissues with particular functions

Linked learning opportunities

Practical work:

- investigate mitosis using a microscope to look at stained cells from onion root tip

Specification links:

- factors that increase the risk of developing cancer (B2.5)

B4.4 How is plant growth controlled? *(separate science only)*

Teaching and learning narrative

Plants are able to respond to their environment in different ways, e.g. phototropism in shoots and gravitropism in roots. These responses are controlled and coordinated by a group of plant hormones called auxins, and increase a plant's chances of survival.

Plants can also respond to environmental factors using other hormones. Gibberellins are involved in breaking seed dormancy (germination) in response to water, and bolting (production of flowers in an attempt to reproduce before death) in response to cold or lack of water. Ethene is involved in the ripening of fruit and dropping of leaves. Humans can exploit these responses and others such as triggering rooting in cuttings, by using plant hormones to trigger responses that are advantageous to us.

Assessable learning outcomes

Learners will be required to:

1. a) explain how plant hormones are important in the control and coordination of plant growth and development, with reference to the role of auxins in phototropisms and gravitropisms
 b) describe practical investigations into the role of auxin in phototropism
 PAG6
2. describe some of the variety of effects of plant hormones, relating to gibberellins and ethene
3. describe some of the different ways in which people use plant hormones to control plant growth

Linked learning opportunities

Practical work:

- investigate phototropism and the role of auxins in seedlings, using directional light sources and foil caps and rings

B4.5 Should we use stem cells to treat damage and disease?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Stem cells offer the potential to treat patients by replacing damaged tissues or cells. But the benefits must be weighed against risks and ethical concerns about the use and destruction of human embryos to collect embryonic stem cells. For these reasons, use of stem cells in research and medicine is subject to government regulation in many countries (IaS4).	1. discuss potential benefits, risks and ethical issues associated with the use of stem cells in medicine

Linked learning opportunities

Specification links:

- **stem cell therapy for neuron damage (B5.6)**

Ideas about Science:

- stem cell therapy as an application of science that could change lives (IaS4)
- risks, benefits and ethical issues associated with use of stem cells in medicine (IaS4)